ISHU RAKESHBHAI SINGH

AI / Automation Engineer · InSAR & Geospatial · Full-Stack Builder

Tbilisi, Georgia · +995 555 114 467 · singhishu2060@gmail.com · ishu-portfolio.vercel.app

ABOUT ME

I build AI-assisted software and the plumbing around it — production web apps, Telegram-driven workflow automations, healthcare platforms, and scientific data pipelines. Comfortable across Python, TypeScript and JavaScript; comfortable on AWS; happiest when something needs to actually ship and stay up. Recently I picked up InSAR / Sentinel-1 SAR processing from scratch and delivered a strict EGMS-L3 equivalent road-subsidence pilot on a 3-year stack — the kind of project that mixes a lot of engineering with quite a bit of careful science.

WHAT I'VE BEEN BUILDING

Haimcore (Next.js AI workspace + Amirani assistant)

AI Software Engineer — freelance

2025 – 2026

Operational AI workspace app built around Amirani, an in-product assistant. I owned the product across three releases (v3 multilingual, v3.1 stability, v4 UX overhaul) and the deploy / rollback pipeline that keeps prod survivable.

- Wrote the Next.js front-end and the Express-style API routes; the workspace itself is folders, drafts, notes, and a few specialised editors (Text, URL with Open-Graph fetch, Image with vision). Auto-save fires on 1.5s idle and on every editor unmount.
- Wired Amirani to Anthropic's SDK directly — lazy key resolution that reads `.env.local` when the shell environment shadows it, plus a forced base URL so it never silently routes to a proxy. A small *insight* route appends AI-generated content to a per-day memory summary so future Amirani calls see it as context.
- Built the legacy-shim pattern after a refactor: anything that used to live at `pages/api/.../foo.js` moves to `lib/legacy/foo.js`, behind a thin v4 wrapper that adds CSRF and session. Each shim is tested against the deploy harness so import-path drift bites at smoke-test, not in production.
- Owned the deploy/rollback story end-to-end — tagged pre-deploy snapshots, source-only bundles, an audit folder per release, and a Postgres rollback instance kept around as a safety net for the v4 cut-over.
- Multilingual support across 47 languages, including aliases and fuzzy lookup in the AI Actions menu so non-English keyboards can still find the right option fast.

Relief Guru (AI avatar video pipeline, pharmacy content)

AI Automation Engineer — freelance

2025 – 2026

A Telegram-driven AI video pipeline for a pharmacy-content operator. Drop a brief, get back a finished avatar video posted to YouTube / LinkedIn / Instagram, with everything tracked. Seven n8n workflows, because trying to do this in one was a debugging nightmare.

- **WF1A — Intake Router.** Telegram webhook into n8n. Admin whitelist by chat ID; slash commands (`/new`, `/avatars`, `/voices`, `/templates`, `/status`, `/last`, etc.) route into the right downstream workflow.
- **WF1B — Template + Script Generator.** OpenAI `text-embedding-3-small` on the brief, query against a Pinecone *templates* namespace (top-10 normally, top-100 if the operator provided a hint), then GPT writes the script against the closest template.
- **WF2A — HeyGen Video Generation.** Defensive payload parser that walks every reasonable nested path for the script and avatar/voice IDs — hard-fails locally with a useful error rather than letting HeyGen return its opaque "Bad request". Idempotency guard via static workflow data, 30-second poll loop, max 40 attempts. Telegram "sticky message" UX so progress edits one message instead of spamming the chat.
- **WF2A↔WF2B — Approval Bridge + Posting Core.** Operator approves in Telegram; native posting to YouTube, LinkedIn, Instagram via Google & LinkedIn Developer Console integrations.
- **WF3 — Tracker Engine.** Every step in every run gets logged with `task_id`, `trace_id`, `state`, and an `embed_text` field that goes back into Pinecone for later retrieval and learning.
- Operationally: the original V1 was one giant workflow and impossible to debug. Splitting it into V4 with explicit single-purpose webhooks made every failure traceable to a single workflow ID.

Jalaram Hospital — AI Hospital Management System (live: shreejalaramhospital.live)

Full-Stack Engineer — freelance

2025 – 2026

An AI-assisted hospital management system for a women's health clinic in India — four portals (Patient, Doctor, Staff, Admin), a tool-calling clinical Copilot, daily AI insights for doctors, and a Gynae/Obstetrics-shaped data model. Deployed and live.

- **Four portals, one backend.** *PatientPortal* — appointments, medical records, prescriptions, reports, cycle tracker. *DoctorDashboard* — today's roster, consultations (vitals, GPAL, pregnancy module, pelvic exam notes), prescriptions and templates, AI insights, Copilot. *StaffDashboard / Ops Center* — live counts of scheduled / completed / missed appointments, pending and overdue follow-ups, reports filed today, call attempts logged. *AdminDashboard* — users, audit log, system-wide metrics.
- **JALARAM SUPER-AI Copilot.** GPT-4o with tool-calling against the Postgres schema — `search_patients` by name or `PAT-1-2026` hospital ID, `get_patient_context` for the full clinical bundle. Per-doctor chat sessions, auto-titled by the model, conversation history persisted to `CopilotSession` / `CopilotMessage`.
- **Doctor AI Insights (GPT-4o).** Every morning the dashboard anonymises the roster — appointments today, follow-up queue, cycle alerts (cycle length below 21 or above 35) — strips PII down to patient IDs only, hands the bundle to GPT-4o, and asks for exactly three actionable bullets. The prompt stays on a strict, HIPAA-flavoured tone.
- **Gynae / Obstetrics-shaped schema.** Pregnancy module (trimester, gestational age, risk category, EDD), GPAL counts (gravida / para / abortions / living children), menstrual cycle logs with flow / symptoms / ovulation and fertile-window prediction, clinical consultations with vitals + chief complaint + pelvic exam notes, operation records (surgeon, assistant, anaesthesia type, complications).
- **Reports + Lab uploads.** Multer-S3 to AWS S3 (`ap-south-1`) with a Reports table that carries an optional AI summary field — the room for the next step, which is auto-summarising uploaded lab/ultrasound reports back into the patient's timeline.
- **Production:** Express 5 + TypeScript + Prisma 6 over PostgreSQL; JWT cookies + bcrypt + helmet; CORS whitelist for the live domain; request logger; 20 MB body limit for report uploads. Docker + Nginx + PM2

on EC2; IST-aware (+5h 30) date helpers everywhere because the dashboards compute “today” in clinic local time.

SATALITE — Sentinel-1 road subsidence monitoring

InSAR / Remote-Sensing Engineer — freelance (Upwork client)

April – May 2026

Two pilot deliverables on a road-subsidence brief: a strict EGMS-L3 equivalent pipeline at Najafgarh Road, SW Delhi (vertical + east-west, ITRF14 frame) and an L2-equivalent single-track pilot at Kite Beach Road, Visakhapatnam. End-to-end open-source pipeline on a local Linux workstation.

- **Delhi — strict EGMS-L3.** 165 Sentinel-1 SLC bursts (88 ASC path 27 IW2 + 77 DSC path 136 IW1), pair-by-pair in SNAP — ESD coregistration, 35×6 multilook to ~80 m, Goldstein, SNAPHU unwrap externalised because SNAP’s internal SNAPHU silently stalled on three pairs. 476 SBAS interferograms, MintPy SBAS inversion with ERA5 tropospheric correction via PyAPS3. ITRF14 absolute frame via Altamimi 2017 Indian-plate Euler pole, MIDAS-validated against LCK4 / HYDE / IISC. True 2D V_U + V_E decomposition per pixel by Cramer’s rule (determinant ~0.95 across the AOI). 2,555 persistent scatterers. V_U −10.3 to +4.7mm/yr; V_E −7.8 to +8.4. Residual RMS ~2mm.
- **Kite Beach — L2-equivalent.** 86 descending SLCs (no ascending coverage at the AOI — Sentinel-1 has a known gap over coastal east India), 244 SBAS pairs retained after coherence filtering, 152 PS at temporal coherence ≥ 0.85. LOS velocity −11.7 to +8.5mm/yr, RMSE ~5mm.
- **Failures and workarounds I lived through:** JVM OOM on SNAP six times before I capped `gpt` at three concurrent pairs; a first multilook batch at 30×5 came out smeared and got rerun at 35×6; ESD coherence threshold dropped from 0.3 to 0.2 on a few problem pairs (acceptable for *this* AOI, wouldn’t generalise); three ERA5 GRIBs came back zero-byte from the CDS endpoint and would silently segfault PyAPS3 three days later if not caught; one pair (`20240623_20240705`) needed explicit `firstBurstIndex / lastBurstIndex` matching the rest of the stack to load cleanly in MintPy.
- **Per-site Streamlit dashboard** — PS map with velocity colour ramp, time-series at the most-subsiding pixel, decomposition QA scatter, SBAS network graph, data explorer. Both dashboards deployed to AWS EC2 for the client.

TECH I LEAN ON

Languages	Python · TypeScript · JavaScript · Java
AI & LLMs	OpenAI APIs (GPT-4o, embeddings, tool calling) · Anthropic SDK · LangChain · Pinecone (vector) · RAG · Prompt engineering
InSAR / Geospatial	Sentinel-1 SAR · SNAP / ESA STEP · MintPy SBAS · SNAPHU · PyAPS3 / ERA5 · ASF HyP3 · ITRF14 / Indian-plate PMM · GeoPandas · GDAL · QGIS · Streamlit
Cloud & ops	AWS (EC2, S3, RDS, IAM, Lambda) · Docker · docker-compose · Nginx · PM2 · Serverless
Backend	Node.js · Express 5 · FastAPI · Next.js (pages) · REST · JWT · bcrypt · helmet
Frontend	React · Vite · TypeScript · Next.js · Streamlit · Plotly
Databases	PostgreSQL · Prisma 6 ORM · MongoDB · Redis · Firebase · Pinecone (vector)
Automation	n8n (multi-workflow architectures) · Telegram Bot API · Make.com · Zapier · Webhooks · Event-driven pipelines

Integrations	OpenAI · Anthropic · HeyGen · Creatomate · Pinecone · Google Workspace APIs · LinkedIn Developer APIs · ASF Vertex · ESA Copernicus · Telegram Bot API
Other	Git · PowerShell · pdfkit / pdf-parse · Multer-S3 · Helmet / CSRF / cookie auth · Agile

TRAINING & CERTIFICATIONS

AI & Machine Learning Specialisation — Generative AI, neural networks, applied ML (Udemy, Coursera, Campus X)

LangChain & Generative AI Development — Production LLM applications and patterns (Educative + industry training)

AWS Cloud Fundamentals & Serverless Architecture — EC2 / S3 / RDS / IAM / Lambda, infrastructure design

Full-Stack Web Development — Node.js + Python + React across multiple frameworks

Workflow Automation & Integration Platforms — Hands-on n8n / Make.com / Zapier across real client work

Healthcare IT Systems Architecture — HMS / EHR data models, HIPAA-aware workflows

Sentinel-1 InSAR Processing — SNAP / MintPy / SNAPHU pipelines, EGMS-L3 methodology

LANGUAGES I SPEAK

English (Fluent) · **Hindi** (Native) · **Gujarati** (Native) · **Georgian** (Conversational)